AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2024

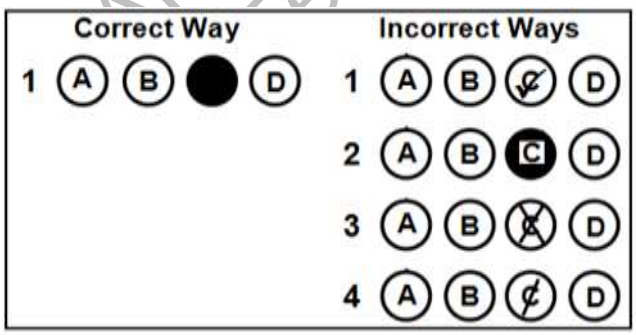
Mathematics Paper I

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. A formulae list is provided on page 2 and 3. You may refer to it during the paper, if you wish.
8. You may use a scientific calculator if you wish.

List of Formulae

Note:

- All symbols used in the formulae have their usual meaning.

Functions and Limits

$$\lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1 \qquad \lim_{x\to 0}\frac{\sqrt{x+a}-\sqrt{a}}{x}=\frac{1}{2\sqrt{a}} \qquad \lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}$$

$$\lim_{x\to 0}(1+x)^{\frac{1}{x}}=e \qquad \lim_{x\to+\infty}\left(1+\frac{1}{x}\right)^x=e \qquad \lim_{x\to 0}\frac{a^x-1}{x}=\log_e a$$

Differentiation

$$\frac{\Delta y}{\Delta x}=\frac{f(x+h)-f(x)}{h} \qquad \frac{d}{dx}(\sec x)=\sec x\tan x \qquad \frac{d}{dx}x^n=nx^{n-1} \qquad \frac{d}{dx}(\sin x)=\cos x$$

$$\frac{d}{dx}(\cos x)=-\sin x \qquad \frac{d}{dx}(\tan x)=\sec^2 x \qquad \frac{d}{dx}(\cot x)=-\operatorname{cosec}^2 x \qquad \frac{d}{dx}(\operatorname{cosec} x)=-\operatorname{cosec} x\cot x$$

$$\frac{d}{dx}\left[\sin^{-1}x\right]=\frac{1}{\sqrt{1-x^2}},\ x\in(-1,1) \qquad \frac{d}{dx}\left[\cos^{-1}x\right]=-\frac{1}{\sqrt{1-x^2}},\ x\in(-1,1) \qquad \frac{d}{dx}\left[\tan^{-1}x\right]=\frac{1}{x^2+1},\ x\in R \qquad \frac{d}{dx}\left[\sec^{-1}x\right]=\frac{1}{|x|\sqrt{x^2-1}},\ x\in[-1,1]'$$

$$\frac{d}{dx}\left[\operatorname{cosec}^{-1}x\right]=-\frac{1}{|x|\sqrt{x^2-1}},\ x\in[-1,1]' \qquad \frac{d}{dx}\left[\cot^{-1}x\right]=-\frac{1}{1+x^2},\ x\in R \qquad \frac{d}{dx}(a^x)=a^x\ln a \qquad \frac{d}{dx}(\log_a x)=\frac{1}{x}\times\frac{1}{\ln a}$$

$$\frac{d}{dx}(\sinh x)=\cosh x \qquad \frac{d}{dx}(\cosh x)=\sinh x \qquad \frac{d}{dx}(\tanh x)=\operatorname{sech}^2 x$$

$$\frac{d}{dx}(\operatorname{cosech} x)=-\coth x\operatorname{cosech} x \qquad \frac{d}{dx}(\operatorname{sech} x)=-\tanh x\operatorname{sech} x \qquad \frac{d}{dx}(\coth x)=-\operatorname{cosech}^2 x$$

$$\frac{d}{dx}[f(x)\times g(x)]=f(x)\frac{d}{dx}[g(x)]+g(x)\frac{d}{dx}[f(x)] \qquad \frac{d}{dx}\left(\frac{f(x)}{g(x)}\right)=\frac{g(x)\frac{d}{dx}[f(x)]-f(x)\frac{d}{dx}[g(x)]}{[g(x)]^2}$$

Maclaurin Series $f(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\frac{f'''(0)}{3!}x^3+\ldots+\frac{f^n(0)}{n!}x^n+\ldots$

Taylor's Series $f(x)=f(a)+f'(a)(x-a)+\frac{f''(a)}{2!}(x-a)^2+\frac{f'''(a)}{3!}(x-a)^3+\frac{f^{iv}(a)}{4!}(x-a)^4+\ldots+\frac{f^n(a)}{n!}(x-a)^n+\ldots$

Integration

$$\int f'(x)dx=f(x)+c \qquad \int[f(x)]^n f'(x)dx=\frac{[f(x)^{n+1}]}{n+1}+c,\quad (n\neq -1) \qquad \int f(x)g'(x)dx=f(x)g(x)-\int f'(x)g(x)dx$$

$$\int\frac{f'(x)}{f(x)}dx=\ln f(x)+c \qquad \int\operatorname{cosec} x\,dx=\ln|\operatorname{cosec}x-\cot x|+c \qquad \int\sec x dx=\ln|\sec x+\tan x|+c$$

Plane Analytical Geometry (Straight Line)

$$d=\frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}} \qquad y-y_1=\frac{y_2-y_1}{x_2-x_1}(x-x_1) \qquad \left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$$

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Point of internal division

$$\left(\frac{k_1x_2+k_2x_1}{k_1+k_2},\ \frac{k_1y_2+k_2y_1}{k_1+k_2}\right)$$

Point of external division

$$\left(\frac{k_1x_2-k_2x_1}{k_1-k_2},\ \frac{k_1y_2-k_2y_1}{k_1-k_2}\right)$$

$$\frac{x-x_1}{\cos\alpha}=\frac{y-y_1}{\sin\alpha}=r$$

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$\frac{x}{a}+\frac{y}{b}=1$$

$$x\cos\alpha+y\sin\alpha=p$$

$$y=mx+c$$

$$y-y_1=m(x-x_1)$$

$$\theta=\tan^{-1}\left[\pm\frac{m_2-m_1}{1+m_1m_2}\right]$$

$$\theta=\tan^{-1}\left[\frac{2\sqrt{h^2-ab}}{a+b}\right]$$

$$\begin{vmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}=0$$

$$\Delta=\frac{1}{2}\begin{vmatrix}x_1 & y_1 & 1\\ x_2 & y_2 & 1\\ x_3 & y_3 & 1\end{vmatrix}$$

Circles

$$(x-h)^2+(y-k)^2=r^2$$

$$x^2+y^2+2gx+2fy+c=0$$

Equation of normal to a circle

$$(y-y_1)(x_1+g)=(x-x_1)(y_1+f)$$

Equation of tangent to a circle

$$xx_1+yy_1+g(x+x_1)+f(y+y_1)+c=0$$

Length of tangent to a circle from a point $(x_1,\ y_1)$, $l=\sqrt{x_1^{\,2}+y_1^{\,2}+2gx_1+2fy_1+c}$

Parabola

$$x^2=4ay \qquad (x-h)^2=4a(y-k) \qquad y^2=4ax \qquad (y-k)^2=4a(x-h)$$

Ellipse

$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1,\ a>b \qquad b^2=a^2(1-e^2) \qquad c=ae \qquad \frac{(y-k)^2}{a^2}+\frac{(x-h)^2}{b^2}=1,\ a>b$$

Hyperbola

$$\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1 \qquad b^2=a^2(e^2-1) \qquad c=ae \qquad \frac{(y-k)^2}{a^2}-\frac{(x-h)^2}{b^2}=1$$

Translation and Rotation

$$x=X+h,\ y=Y+k \qquad X=x\cos\theta+y\sin\theta,\ Y=y\cos\theta-x\sin\theta \qquad \tan 2\theta=\frac{2h}{a-b}$$

Vectors

$$\underline{u}.\underline{v}=|\underline{u}||\underline{v}|\cos\theta \qquad \text{Area of a triangle }=\frac{1}{2}|\underline{u}\times\underline{v}| \qquad \underline{u}\times\underline{v}=\left(|\underline{u}||\underline{v}|\sin\theta\right)\hat{\underline{n}} \qquad \underline{r}=\frac{q\underline{a}+p\underline{b}}{p+q}$$

$$\left|\overrightarrow{P_1P_2}\right|=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2} \qquad \frac{1}{6}(\underline{u}\times\underline{v}).\underline{w}=\frac{1}{6}[\underline{u}\ \underline{v}\ \underline{w}] \qquad \underline{u}.(\underline{v}\times\underline{w})=\begin{vmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}$$

Some Trigonometric Identities

$$\cos(\alpha\pm\beta)=\cos\alpha\cos\beta\mp\sin\alpha\sin\beta$$

$$\sin(\alpha\pm\beta)=\sin\alpha\cos\beta\pm\cos\alpha\sin\beta$$

$$\cos P+\cos Q=2\cos\frac{P+Q}{2}\cos\frac{P-Q}{2}$$

$$\cos P-\cos Q=-2\sin\frac{P+Q}{2}\sin\frac{P-Q}{2}$$

$$\sin P+\sin Q=2\sin\frac{P+Q}{2}\cos\frac{P-Q}{2}$$

$$\sin P-\sin Q=2\cos\frac{P+Q}{2}\sin\frac{P-Q}{2}$$

$$\sin 2\alpha=2\sin\alpha\cos\alpha$$

$$\cos 2\alpha=\cos^2\alpha-\sin^2\alpha$$

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1. If $x \in R$, then the range of $y = -|x-2|$ will be the set of

   A. real numbers.
   B. positive real numbers.
   C. real numbers except 0.
   D. negative real numbers.

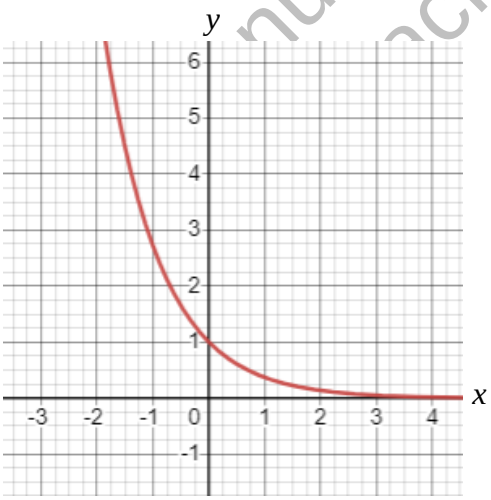
2. The inverse of the function $f(x) = \dfrac{2x+1}{3}$ will be

   A. $\dfrac{3}{2x+1}$.
   B. $-\dfrac{2x+1}{3}$.
   C. $\dfrac{3x-1}{2}$.
   D. $\dfrac{3x+1}{2}$.

3. The left hand limit of the function $f(x) = \begin{cases} 2x+3 & \text{when } x \geq 3 \\ 3x-2 & \text{when } x < 3 \end{cases}$ will be

   A. 3
   B. 6
   C. 7
   D. 9

4. A function $f(x)$ is represented by the given graph.



   The range of $f(x)$ will be

   A. $(-\infty, \infty)$.
   B. $(-\infty, 0)$.
   C. $(0, \infty)$.
   D. $(1, \infty)$.

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5. The expression $\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ represents

I. derivative
II. average rate of change
III. instantaneous rate of change

A. I only.
B. II only.
C. I and III.
D. II and III.

6. The derivative of $y = a^e \times x^e$, with respect to *x*, will be

A. 0
B. $ea^{e-1} \times ex^{e-1}$
C. $a^e \times ex^{e-1}$
D. $ea^{e-1} \times x^e + a^e \times ex^{e-1}$

7. The derivative of $x(x-2)^5$, with respect to *x*, is

A. $5(x-2)^4$
B. $5x(x-2)^4$
C. $5(x-2)^4 + (x-2)^5$
D. $5x(x-2)^4 + (x-2)^5$

8. The derivative of the function $xy - 1 = 0$, with respect to *x*, will be

A. $-\frac{x}{y}$.
B. $-\frac{1}{x^2}$.
C. $\frac{x}{y}$.
D. $\frac{1}{x^2}$.

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9. If $y = 2^{\ln x}$, then $\frac{dy}{dx}$ will be equal to

A. $\frac{2^{\ln x}}{x}$.

B. $\frac{2^{\ln x} \times \ln 2}{x}$.

C. $2^{\ln x} \times \ln 2$.

D. $2^{\ln x} \times \ln x$.

10. If $f(x) = \sinh(\pi - x)$, then $f'(x)$ will be

A. $-\cosh(\pi - x)$.

B. $\cosh(\pi - x)$.

C. $-(\pi - x)\cosh(\pi - x)$.

D. $(\pi - x)\cosh(\pi - x)$.

11. The slope of the tangent to the curve $y = \frac{7}{4}x + \frac{3}{x}$ at $x = 2$ is

A. 0

B. 1

C. $\frac{5}{2}$

D. $\frac{3}{4}$

12. The function $f(x) = x^2 - 2x - 3$ has an extreme value at

A. $x = -1$

B. $x = 1$

C. $x = 2$

D. $x = 3$

13. To convert $\frac{1}{(x^2 + a)(x^2 - b^2)}$ into its partial fractions, the appropriate form will be

A. $\frac{Ax + B}{(x^2 + a)} + \frac{Cx + D}{x^2 - b^2}$.

B. $\frac{Ax + B}{(x^2 + a)} + \frac{C}{x^2 - b^2}$.

C. $\frac{Ax + B}{(x^2 + a)} + \frac{C}{x - b} + \frac{D}{x + b}$.

D. $\frac{Ax + B}{(x^2 + a)} + \frac{C}{x - b} + \frac{D}{x^2 - b^2}$.

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14. The integral $\frac{d}{dx}\int \cos(\sin x)\,dx$ is equal to

A. $\cos(\sin x)$.
B. $\sin(\sin x)$.
C. $-\sin(\sin x)$.
D. $-\cos(\sin x)$.

15. If $f'(x) = \frac{\ln x}{x}$, then $f(x)$ will be

A. $\ln(\ln x) + C$.
B. $\frac{(\ln x)^2}{2} + C$.
C. $\frac{(\ln x)^2}{2x} + C$.
D. $\ln x + C$.

16. $\int \ln(\sqrt{x}) \times \frac{1}{\sqrt{x}}\,dx$ is equal to

A. $2\left[\ln(\sqrt{x})\right]^2 + C$.
B. $\left[\ln(\sqrt{x})\right]^2 + C$.
C. $\frac{\ln x}{2\sqrt{x}} + C$.
D. $\frac{\ln x}{\sqrt{x}} + C$.

17. If $\int xe^x dx = xe^x - \int p(x)dx$, then the expression $p(x)$ is equal to

A. 1
B. x
C. e^x
D. xe^x

18. If $\int_0^1 (4x + k)\,dx = 8$, then the value of k is

A. 0
B. 6
C. 8
D. 10

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19. The order of the differential equation $x^3 \frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 0$ is

A. 0
B. 1
C. 2
D. 3

20. If a function is an odd function, then the value of $\int_{-\pi}^{\pi} f(x)\,dx$ will be

A. $-\pi$
B. 0
C. π
D. 2π

21. The point (1, x) divides the line segment joining (–2, –1) and (2, 3) in the ratio of 3:1.

The value of x will be

A. -2
B. -1
C. 1
D. 2

22. Which of the given equations represents the line perpendicular to $y = 2x + 3$?

A. $y = -2x + 3$
B. $y = -\frac{1}{2}x + 3$
C. $y = -x + 3$
D. $2y = 4x + 6$

23. The equation of the line passing through (a, b) and perpendicular to x–axis is

(**Note**: Here a and b are positive real numbers.)

A. $x + a = 0$
B. $x - a = 0$
C. $y - a = 0$
D. $y + a = 0$

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24. In normal form, $2x - y = 1$ can be written as

A. $\frac{2x}{3} - \frac{y}{3} = \frac{1}{3}$.

B. $\frac{2x}{5} - \frac{y}{5} = \frac{1}{5}$.

C. $\frac{2x}{\sqrt{3}} - \frac{y}{\sqrt{3}} = \frac{1}{\sqrt{3}}$.

D. $\frac{2x}{\sqrt{5}} - \frac{y}{\sqrt{5}} = \frac{1}{\sqrt{5}}$.

25. Which of the given points lie above the line $x - 2y - 1 = 0$?

A. $(0, -1)$
B. $(-4, -1)$
C. $(1, 0)$
D. $(1, 4)$

26. The distance between two parallel lines $x - y - 1 = 0$ and $x - y - 2 = 0$ is

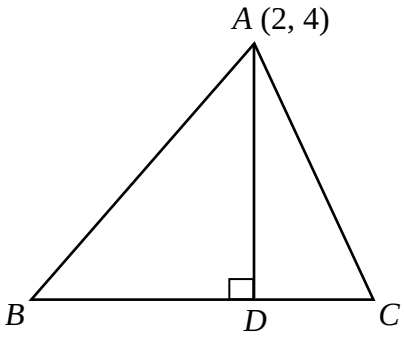
A. 1 unit.
B. 2 units.
C. $\frac{1}{\sqrt{2}}$ units.
D. $\sqrt{2}$ units.

27. The angle from the line $x - \frac{y}{2} = 2$ to the line $2x - y = 2$ is

A. 0°
B. 45°
C. 90°
D. undefined

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28. In the given triangle *ABC*, *AD* is one of the altitudes from *A* to *BC*.



NOT TO SCALE

If the slope of *BC* is $\frac{2}{3}$, then the equation of *AD* will be

A. $y = \frac{2}{3}x + 7$

B. $y = -\frac{3}{2}x + 7$

C. $y = \frac{2}{3}(x - 2) + 7$

D. $y = -\frac{3}{2}(x - 2) + 7$

29. The equations for two straight lines passing through the origin on a Cartesian plane are $x + 2y = 0$ and $y = x$.

The joint equation for the two given lines will be

A. $2x + 3y = 0$

B. $x^2 + 2y^2 = 0$

C. $xy + 2y^2 = 0$

D. $x^2 + xy - 2y^2 = 0$

30. If the lines $y = 2$, $x = b$ and $x = -y$ are concurrent, then the value of *b* will be

A. -2

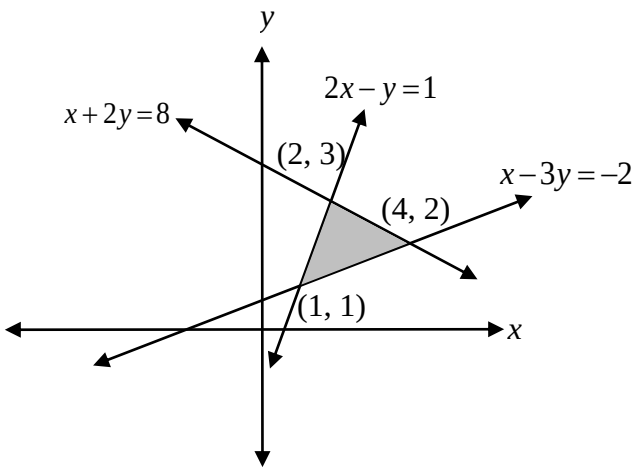
B. -1

C. 1

D. 2

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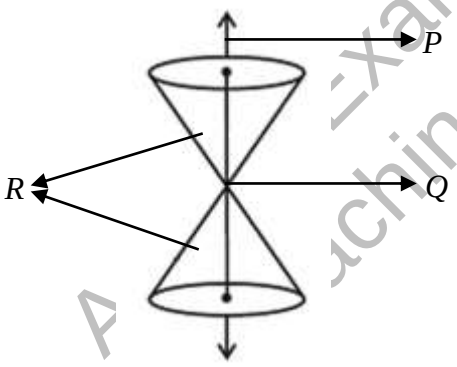
31. The graphic solution of a linear programing problem is presented in the given graph.



Based on the given information, one of the constraints of the feasible region is

A. $x < y < 0$
B. $x - 3y \geq -2$
C. $2x - y \leq 1$
D. $x + 2y \leq 8$

32. Consider the given right cone.



Which of the following options shows the CORRECT identification of labels on the given cone?

	P	***Q***	***R***
A	Axis	Nappe	Vertex
B	Vertex	Nappe	Axis
C	Vertex	Axis	Nappe
D	Axis	Vertex	Nappe

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33. The centre of the circle $x^2 + y^2 + 8x + 18y + 5 = 0$ is

A. (4, 9).
B. (8, 18).
C. (– 4, – 9).
D. (– 8, –18).

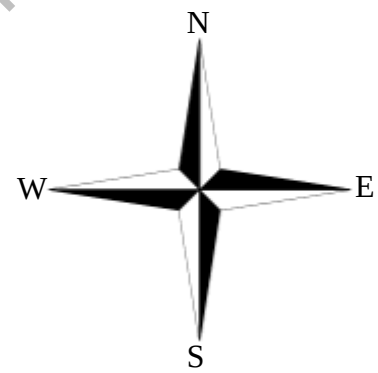
34. The line $y = 2x + p$ is tangent to the circle $x^2 + y^2 = 16$ and it satisfies the condition $a^2(1 + m^2) - c^2 = 0$.

In the given statement, the value of *p* is equals to

A. 2
B. 4
C. $\pm 4\sqrt{5}$
D. $\pm 16\sqrt{5}$

35. A cell phone transmission tower provides service to its subscribers within a range (radius) of 10 km. The following people are located at the given positions with respect to the tower.

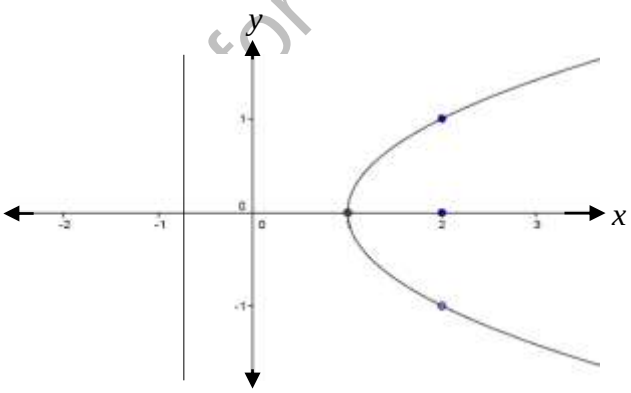
- Hamid is 9 km east and 5 km south
- Sohail is 8 km west and 8 km south
- Aliza is 8 km east and 8 km north
- Sana is 7 km west and 6 km north



Which of the subscriber will be able to receive the service?

A. Hamid
B. Sohail
C. Aliza
D. Sana

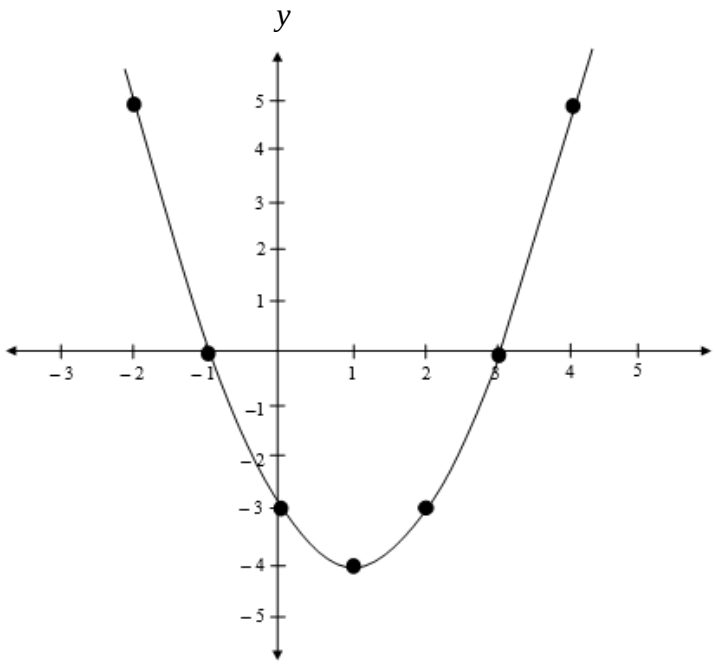
36. In the given parabola, the length of latus rectum is



A. 1 unit.
B. 2 unit
C. 3 unit.
D. 4 unit.

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Use the given graph to answer Q.37 and Q.38.



37. The equation for the given parabola will be

A. $(x-1)^2 = y+4$
B. $(y-1)^2 = x+4$
C. $(x+1)^2 = y-4$
D. $(y+1)^2 = x-4$

38. If the focus of the given parabola is $\left(1, -\frac{15}{4}\right)$, then the equation for the directrix will be

A. $x = \frac{1}{4}$.
B. $y = \frac{1}{4}$.
C. $x = -\frac{17}{4}$.
D. $y = -\frac{17}{4}$.

39. If an ellipse is represented by the equation $\frac{x^2}{4} + y^2 - 4y + 4 = 1$, then its centre would lie at

A. $(0, 2)$.
B. $(2, 0)$.
C. $(-2, 0)$.
D. $(0, -2)$.

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40. A garden has an elliptic walking track around it. On the walking track, the maximum distance is 20 m. Assume that, the minimum distance between a person sitting at the centre and a person standing on the walking track is 6 m.



The equation that represents the elliptic path will be

A. $\frac{x^2}{100}+\frac{y^2}{6}=1$

B. $\frac{x^2}{200}+\frac{y^2}{36}=1$

C. $\frac{x^2}{100}+\frac{y^2}{36}=1$

D. $\frac{x^2}{200}+\frac{y^2}{6}=1$

41. If the equation of a hyperbola is $\frac{x^2}{16}-\frac{y^2}{25}=1$, then equation for the asymptotes will be

A. $x=\frac{16}{25}y$.

B. $y=\frac{25}{16}x$.

C. $x=\pm\frac{4}{5}y$.

D. $y=\pm\frac{5}{4}x$.

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42. The slope of tangent to a hyperbola at the point (x, y) is *m*. The equation of line which is perpendicular to this tangent at (x, y) will be

(**Note:** *c* is a constant.)

A. $mx + y = c$.
B. $mx - y = c$.
C. $x + my = c$.
D. $x - my = c$.

43. The transverse axis of a hyperbola is along *x*-axis with centre at (0, 0). The distance between its foci is $2\sqrt{41}$. If the length of the transverse axis is 10, then the equation of this hyperbola will be

A. $\frac{x^2}{9} - \frac{y^2}{25} = 1$
B. $\frac{x^2}{25} - \frac{y^2}{9} = 1$
C. $\frac{x^2}{25} - \frac{y^2}{16} = 1$
D. $\frac{x^2}{16} - \frac{y^2}{25} = 1$

44. Consider a point (2, 3) on a Cartesian plane. If the axis is translated 5 units towards positive *x*-axis and 2 units along positive *y*-axis, then the point with respect to the new origin will be

A. (7, 5).
B. (4, 8).
C. (− 3, 1).
D. (0, − 2).

45. If $\hat{i} - a\,\hat{j}$ is a unit vector, then the value of *a* will be

A. 0
B. 1
C. $\sqrt{2}$
D. 2

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46. If the vectors $\vec{u} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{v} = \hat{j} - b\hat{k}$ are orthogonal vectors, then the value of *b* will be

A. $-\frac{1}{2}$
B. 0
C. $\frac{1}{2}$
D. 1

47. If $a = 2\hat{i} - \hat{j}$ and $b = x\hat{i} - 3\hat{j}$ are two parallel vectors, then the value of *x* will be

A. 2
B. 3
C. 4
D. 6

48. The scalar triple product of vectors $a = \hat{i} + 2\hat{j}$, $b = \hat{i} - \hat{j}$ and $c = \hat{i} + \hat{k}$ is

A. -1
B. -3
C. 0
D. 2

49. If $a = 2\hat{i} - \hat{j}$ and $b = \hat{i} - \hat{j}$ are two vectors, then projection of vector ***a*** on vector ***b*** will be

A. $\frac{3}{\sqrt{2}}$.
B. $\frac{3}{2}$.
C. $\frac{3}{\sqrt{5}}$.
D. $\frac{3}{5}$.

50. The dot product of two vectors $\sqrt{3}\,\hat{i} - \sqrt{3}\,\hat{j}$ and $\sqrt{3}\,\hat{i} + \sqrt{3}\,\hat{k}$ is

A. 0
B. $\sqrt{3}$
C. $2\sqrt{3}$
D. 3

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